

Anomaly detection in streaming data with gaussian process based stochastic differential equations

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ABSTRACT

This paper characterises streaming data as the evolution of a stochastic differential equation, with the aim of extracting information that can be used to detect anomalies in the stream. Gaussian process regression provides a flexible approach to approximating components of the stochastic differential equation, allowing for complex modelling of underlying data generation dynamics. The proposed algorithm displays superior discriminative power over different time-series anomaly detection methods for both synthetic and NYC taxi datasets, whilst the introduced bootstrapping method for setting the detection threshold provides control over the false alarm rate of the anomaly detector.

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1. Introduction

Anomaly detection is a key application of streaming data analysis, with examples including quality control, system health monitoring to fraud and intrusion detection. Anomaly detection systems automatically detect anomalous data, and alert human analysts of the problem so that it can be resolved. The improvement of anomaly detection systems can prevent unnecessary human interventions by reducing false positive rates [1], and allow analysts to react quicker to developing problems by reducing detection times.

In [2], a general definition of an anomaly is given as “the measurable consequences of an unexpected change in state of a system which is outside of its local or global norm”. Types of anomalies include Global (a.k.a. point), Collective and Contextual anomalies, detailed in [3]. Detrending and deseasonalizing the data to give the residual process as a preprocessing step provides a way to detect all these types of anomalies. Anomalies appear as a deviation from the normal, and decomposing the data in this way implies a normal behaviour of seasonality and trend, and so anomalous data would be entirely contained within this residual process.

The main purpose of this paper is to combine nonparametric estimation techniques for stochastic differential equations (SDEs), with statistical testing methodology to provide a general algorithm for streaming anomaly detection. SDE estimation is used to extract

both the deterministic and stochastic dynamics of the normal data generation process, and statistical testing used to compare incoming data to this learned model to test for the presence of anomalies.

2. Related works

The nonparametric estimation of SDEs has been used to characterise and distinguish between different data generation processes in many applications. Initial work on SDE estimation focused on learning the dynamics of a system based on noisy state observations [4,5]. In [4], histogram based learning of the deterministic and stochastic parts of the SDE was applied to differentiating between classes of patient tremors, using neurophysiological time-series data. In [5], human heartbeat dynamics were studied for healthy patients and patients with congestive heart failure. The drift and diffusion coefficients showed significantly different structures, and could be used to distinguish between healthy and ill patients. Gaussian process (GP) based classifiers have also been used for distinguishing between patients based on streaming medical data [6], outperforming non-probabilistic methods.

Further work improved upon the histogram based estimation methods by using kernel-based regression to learn drift and diffusion coefficient, with reconstruction based cross-validation (CV) for setting kernel bandwidth [7]. In [8], local polynomial regression has been used to estimate the coefficients, with reconstruction based CV to select the optimal polynomial orders. This paper also provides an automatic method for comparing different SDE

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models to detect anomalies, based on a likelihood ratio formed directly from drift and diffusion coefficients. This paper seeks to extend this methodology to include more flexible function estimation, model validation, and to improve upon threshold setting techniques.

More recent work has focused on Gaussian Process Regression (GPR) based estimation of drift and diffusion coefficients [9], and also Bayesian methods of estimation [10]. GPR allows for improved hyperparameter setting through marginal-likelihood optimisation, detailed in [9]. In [11], SDE estimation is again cast as a GPR problem, and set up in state-space form to allow for efficient Kalman filtering to be used, but is limited to estimation of the drift function only. For sparse observations, [11] presents high-order Itô–Taylor expansions of transition densities, over the standard Euler–Maruyama scheme, and [12] imputes data points between time points.

Automated anomaly detection requires two components, an anomaly score, and a threshold for detection. Deep learning methods often predict incoming data by training a neural network on normal data, using the prediction errors as anomaly scores [3]. Statistical methods provide probabilistic models for data, which can use likelihood ratios for anomaly scores [8,13].

Setting a threshold for detection in an unsupervised context provides a challenge which is addressed in several different ways in the literature. In [14], time-series prediction errors are approximated by a Gaussian, and decision threshold was set based on the desired quantile of the approximating Gaussian cumulative distribution function (CDF). When the Gaussian prediction assumption is dropped, the histogram of prediction errors can be used to adaptively set the threshold based on empirical quantile estimates [15]. In [13], a Monte-Carlo approach was used to set the threshold based on a generalised likelihood ratio (GLR) test statistic, however, this threshold does not directly relate to control of the false alarm rate of the classifier.

3. Methodology

In this section, we outline the relevant theory and extensions that allow us to perform anomaly detection using a GPR based estimate of an SDE. Firstly, SDE estimation using GPR is outlined in Section 3.1, which is followed by the validation method used to set hyperparameters. We apply stochastic process theory to the GP based SDEs to determine a test statistic used for the anomaly detection. We use bootstrapping methodology for setting the decision threshold of the statistical test to control the false alarm rate of the detector.

3.1. Stochastic differential equation estimation

A univariate autonomous SDE is used to model the data generating process. The form of the SDE is given in Eq. (1), where the functions $f(\cdot)$, $D(\cdot)$ are the commonly termed *drift* and *diffusion* coefficients.

$$dX_t = f(X_t)dt + \sqrt{D(X_t)}dW_t \quad (1)$$

where W_t is a standard Wiener process, and $W_{t+\Delta t} - W_t \sim N(0, \Delta t)$. Nonparametric estimation of these coefficients relies on the small-time limit of the conditional expectations, Eqs. (2), (3).

$$f(x) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \mathbb{E}[X_{t+\Delta t} - X_t | X_t = x] \quad (2)$$

$$D(x) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \mathbb{E}[(X_{t+\Delta t} - X_t)^2 | X_t = x] \quad (3)$$

Assuming dense observations (small Δt), the following approximation to the drift and diffusion functions are given by the corresponding expectations.

$$f(x) \approx \mathbb{E}\left[\frac{X_{t+\Delta t} - X_t}{\Delta t} \middle| X_t = x\right] \quad (4)$$

$$D(x) \approx \mathbb{E}\left[\left(\frac{X_{t+\Delta t} - X_t}{\sqrt{\Delta t}}\right)^2 \middle| X_t = x\right] \quad (5)$$

Using the Euler–Maruyama discretization [16], we approximate state transitions with a Gaussian distribution, which discards higher-order moments of the true transition distribution, assuming that terms $\mathcal{O}(\Delta t^2)$ are negligible. Defining $\Delta X_{t_i} = X_{t_i+\Delta t} - X_{t_i}$, where $t_i \in [0, T]$, $t_{i-1} < t_i < t_{i+1}$ and $\frac{T}{N} = \Delta t$, where T is the final time and N is the total number of time steps.

$$\Delta X_{t_i} \approx f(X_{t_i})\Delta t + \sqrt{D(X_{t_i})}\epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \Delta t) \quad (6)$$

Once the diffusion function has been learned, this approximation produces a Gaussian likelihood for the drift function, and so placing a conjugate GP prior over the drift functions allows for exact Bayesian inference to produce a GP posterior over the drift function. This is detailed in Section 3.1.2, and [10].

3.1.1. Diffusion estimation

GPR is used to approximate these expectations [10], and so now we formulate the diffusion estimation as a typical regression problem. We define the training point/data point pairs $(x_i, \tilde{y}_i)$, with the data $\tilde{y}_i$ defined as the realisations of the random variables $\tilde{y}_i = (X_{t_i+\Delta t} - X_{t_i})^2 / \Delta t$.

These data points are modelled as noisy function values $\tilde{y}_i = D(x_i) + \zeta_i$, with the diffusion function $D(\cdot)$, and ζ_i a zero-mean random variable, that is not necessarily Gaussian distributed. In this sense, the GPR is being used as a tool for calculating the expectation in Eq. (5), and not for exact Bayesian inference. See Section 5 for a demonstration of GPR's advantage over other non-parametric regression techniques, i.e. polynomial regression. A GP prior is placed on the diffusion function $D(x) \sim \mathcal{GP}(0, k_D(x, x'))$, with a covariance function defined by the kernel $k_D(\cdot, \cdot)$. Note that the diffusion function has a non-zero probability of taking negative values due to the GP taking values on the whole real line, which is problematic as the diffusion is a positive quantity. If this does occur during evaluation, the absolute value is taken. A more natural way to model a positive quantity such as diffusion would be by modelling a function $Q(x)$ with a GP, and transforming using a strictly increasing function $\phi: \mathbb{R} \rightarrow (0, \infty)$ such as the exponential function. However, a significant bias is introduced when finding expectations due to Jensen's inequality: $\mathbb{E}[D(x)] = \mathbb{E}[\phi(Q(x))] \geq \phi(\mathbb{E}[Q(x)])$, with the bias known as “Jensen's gap”.

Defining the vector $\mathbf{k}_D(x)$, as the covariances between the test point x , and the training points as follows

$$\mathbf{k}_D(x) = [k_D(x, x_0) \quad k_D(x, x_1) \quad \dots \quad k_D(x, x_{N-1})]^T \quad (7)$$

with the data point vector as

$$\tilde{\mathbf{y}} = [\tilde{y}_0 \quad \tilde{y}_1 \quad \dots \quad \tilde{y}_{N-1}]^T \quad (8)$$

and the Gram matrix K_D , which is found by evaluating the covariance function between pairs of training points $[K_D]_{ij} = k_D(x_i, x_j)$.

Setting the noise variances for the diffusion regression with a parameter λ , such that $\mathbb{E}[\zeta_i^2] = \lambda$, provides a regularization term in the regression. The resulting mean estimate is given by $\hat{D}(x) = \mathbf{k}_D(x)^T [K_D + \lambda I]^{-1} \tilde{\mathbf{y}}$ which is then passed to the drift regression to condition on.

3.1.2. Drift estimation

Using the estimated diffusion at the training points as the noise variances for the drift GPR, provides a fully Bayesian estimate for the drift function, which gives a posterior mean function estimate as well as posterior variances. Again, defining the regression as noisy drift function values with training/data point pairs (x_i, y_i) , with the data points y_i as the realisations of the random variables $Y_i = (X_{t_i+\Delta t} - X_{t_i})/\Delta t$. Modelling the data points as noisy function values gives $Y_i = f(x_i) + v_i$.

Now from the Euler–Maruyama discretization, Eq. (6), we assumed the noise variances are Gaussian distributed as $v_i \sim \mathcal{N}(0, \hat{D}(x_i)/\Delta t)$. We write this compactly by defining the covariance matrix of the noise as $\Sigma = \frac{D}{\Delta t}$, with

$$D = \text{diag}[\hat{D}(x_0) \quad \hat{D}(x_1) \quad \dots \quad \hat{D}(x_{N-1})]$$

as the diagonal matrix of diffusion estimates at the training points.

The GP prior over the drift functions is defined using the covariance function $k(\cdot, \cdot)$, giving $f(x) \sim \mathcal{GP}(0, k(x, x'))$. The Bayesian posterior can be found analytically, and is given by the Gaussian distribution in Eq. (9), where the vectors $\mathbf{k}(x)$ and $\mathbf{y}$ are constructed the same as Eqs. (7) and (8) respectively, with the corresponding drift kernel $k(\cdot, \cdot)$ and drift data points y_i .

$$\hat{f}(x) \sim \mathcal{N}(\hat{m}(x), \hat{C}(x)) \quad (9)$$

$$\hat{m}(x) = \mathbf{k}(x)^T [K + \Sigma]^{-1} \mathbf{y} \quad (10)$$

$$\hat{C}(x) = k(x, x) - \mathbf{k}(x)^T [K + \Sigma]^{-1} \mathbf{k}(x) \quad (11)$$

The state dependent drift and diffusion functions are now approximated by the functions $\hat{m}(x), \hat{D}(x)$, where the variance of the drift estimate is given by $\hat{C}(x)$. This variance provides confidence intervals for the drift estimate, and is required for evaluating the Marginal Likelihood of the data.

3.1.3. Choice of kernel

In order to estimate the drift and diffusion coefficients, we must select appropriate kernels to define the drift and diffusion GP covariance functions. The form of the covariance $k(x, x')$ determines the space of functions which the GPR can use to approximate the true underlying function. For example, the length-scale parameter l of the squared exponential (SE) kernel function determines the smoothness of the approximating functions, and the scale parameter s determines the typical magnitude of variation. The SE kernel is given by $k_{SE}(x, x') = s^2 \exp(-\frac{|x-x'|^2}{2l^2})$. Another example is the polynomial kernel, where the degree p determines the polynomial order of the approximating functions. The polynomial kernel is given by $k_{poly}(x, x') = s^2(1 + xx')^p$. This kernel would be useful if it is assumed the data generation dynamics follow one of many popular parametric SDE models [16] that have polynomial functions of the state for the drift and diffusion. The SE kernel is useful in the setting where little is known about the underlying data generation dynamics (e.g. streaming data) because the SE kernel enforces a smoothness of approximating function, without imposing strict conditions on the particular form and so is flexible when estimating functions that are unknown and potentially complex.

3.2. Validation

The kernel hyperparameters for both the drift and diffusion kernels must be set to provide the best resulting SDE model based on the observed data. A simple approach to this is time series cross-validation, seen for [7,8,10]. Subsets of the training data are used to learn the SDE functions, then forecasts are made by simulating from the learned SDE, and are then compared to the true values

to test for similarity. Here however, we have access to a less computationally expensive validation that can be performed with the GPR framework by optimising the marginal likelihood [17] which is performed in the context of SDE estimation in [10]. The marginal likelihood refers to the likelihood of the data, where the functions values have been marginalised out, shown here for data points $\mathbf{y}$, training points $\mathbf{x}$, and drift function $\mathbf{f}$.

$$p(\mathbf{y}|\mathbf{x}, \Sigma) = \int \underbrace{p(\mathbf{y}|\mathbf{f}, \mathbf{x}, \Sigma)}_{\text{likelihood}} \underbrace{p(\mathbf{f}|\mathbf{x}, \Sigma)}_{\text{GP prior}} d\mathbf{f} \quad (12)$$

The marginal likelihood can be found analytically, and is given below for the training points $\mathbf{x}$, data vector $\mathbf{y}$, up to an additive constant.

$$\log p(\mathbf{y}|\mathbf{x}, \Sigma) = - \underbrace{\frac{1}{2} \mathbf{y}^T [K + \Sigma]^{-1} \mathbf{y}}_{\text{Data Fit}} - \underbrace{\frac{1}{2} \log |K + \Sigma|}_{\text{Complexity}} \quad (13)$$

Setting hyperparameters through marginal likelihood optimisation is motivated by the automatic trade off between the data fit and the complexity of the model. Additionally, the marginal likelihood comes at low extra computational cost ($\mathcal{O}(N^2)$) [17], compared to the expensive operation of inverting the $N \times N$ covariance matrix ($\mathcal{O}(N^3)$) that is required for fitting the drift mean and variance (with N as the number of training points). Maximisation of the marginal likelihood can be performed using numerical optimisation techniques, in our case a simple grid search was used as we was observed experimentally that the performance was not over-sensitive to the initial setting of hyperparameters.

3.3. Statistical testing

This section builds the framework for generating a relevant test statistic based on a likelihood ratio test for the presence of anomalies. We also provide a method for controlling the false alarm rate of the detector.

3.3.1. Lamperti transform

Calculating the log-likelihood ratio (LLR) using the drift and diffusion functions requires the Lamperti transform [18], which finds an equivalent unitary diffusion SDE from the full drift and diffusion model. This is done to allow for comparison of two unitary diffusion SDE models through Girsanov's theorem, which is the change-of-measure for stochastic processes [8].

Consider again the univariate autonomous SDE model.

$$dX_t = f(X_t)dt + \sigma(X_t)dW_t \quad (14)$$

The Lamperti transform defines the new process Y_t as a function of the process X_t . The process Y_t solves the following unitary diffusion SDE.

$$Y_t = F(X_t) = \int_z^{X_t} \frac{1}{\sigma(u)} du, \quad dY_t = \mu(Y_t)dt + dW_t \quad (15)$$

where z is an arbitrary value in the state space X_t , and the transformed drift is given by:

$$\mu(Y_t) = \frac{f(X_t)}{\sigma(X_t)} - \frac{1}{2} \frac{\partial \sigma(X_t)}{\partial X_t} \quad (16)$$

To find the transformed drift function, we must find the derivative of $\sigma(x) = \sqrt{D(x)}$, based on our GPR estimate of the diffusion coefficient.

$$\frac{\partial \hat{\sigma}(x)}{\partial x} = \frac{1}{2\hat{\sigma}(x)} \mathbf{k}'_D(x)^T [K_D + \lambda I]^{-1} \tilde{\mathbf{y}} \quad (17)$$

where

$$\mathbf{k}'_D(x) = \left[\frac{\partial}{\partial x} k(x, x_0) \quad \frac{\partial}{\partial x} k(x, x_1) \quad \dots \quad \frac{\partial}{\partial x} k(x, x_{N-1}) \right]^T \quad (18)$$

Now for each kernel choice we must derive the term $\frac{\partial}{\partial x} k(x, x')$. The polynomial kernel and squared exponential kernel derivatives are available analytically, and trivial to compute.

3.3.2. Girsanov's theorem

Defining the following unitary diffusion SDEs.

$$dX_t = f_1(X_t)dt + dW_t, \quad 0 \leq t \leq T \quad (19)$$

$$dX_t = f_2(X_t)dt + dW_t, \quad 0 \leq t \leq T \quad (20)$$

For the discrete set of observations of the process $X_{0:T}$, we can approximate the log-likelihood ratio between these two models using Girsanov's theorem [18].

$$\begin{aligned} \zeta(X_{0:T}) = & \sum_{i=0}^{N-1} [f_2(X_{t_i}) - f_1(X_{t_i})](X_{t_i+\Delta t} - X_{t_i}) \\ & - \frac{1}{2} \sum_{i=0}^{N-1} [f_2^2(X_{t_i}) - f_1^2(X_{t_i})] \Delta t \end{aligned} \quad (21)$$

3.3.3. Hypothesis testing

Now that the likelihood ratio between two models can be calculated for a set of observations $x_{0:T}$, it must be applied in a way that produces a useful test statistic for the presence of an anomaly. Here, we will assume we have access to 'normal' training data, i.e. data with no anomalies present, denoted $x_{0:K}$. This data is used to form a normal model, $(\hat{f}_{\text{train}}, \hat{D}_{\text{train}})$, which can be converted into a unitary diffusion SDE model, with drift $\hat{\mu}_{\text{train}}$.

Test data ($x_{K+1:M}$) can then be compared to the training model to test for anomalies. At each time step, a test model $(\hat{f}_{\text{test}}, \hat{D}_{\text{test}})$ is constructed and transformed to a unitary diffusion model with drift $\hat{\mu}_{\text{test}}$. The data used for learning the i^{th} test model is from a sliding window of W previous data points $\{x_j\}_{j=i-W+1}^i$, where W should be great enough for the SDE estimation to capture the local dynamics of the data generation process.

The hypothesis test computes the likelihood ratio between training and test models for the test data points, and compares this to the detection threshold. The null hypothesis, $\mathcal{H}_0$, assumes the test data points are generated from the same process as the training data, hence the probability density function (PDF) for the training and test model should be similar, and therefore the LLR close to zero. The alternative hypothesis, $\mathcal{H}_1$, assumes the test data points are anomalous, and so the test model which has been learned using anomalous samples should have a higher PDF than the training model.

$$\mathcal{H}_0 : P_{\text{test}}(x_{i-W+1:i}) = P_{\text{train}}(x_{i-W+1:i}) \quad (22)$$

$$\mathcal{H}_1 : P_{\text{test}}(x_{i-W+1:i}) \neq P_{\text{train}}(x_{i-W+1:i}) \quad (23)$$

The log-likelihood ratio is defined as

$$\zeta_i \triangleq \zeta(x_{i-W+1:i}) = \sum_{j=i-W+1}^i \log \frac{P_{\text{test}}(x_j)}{P_{\text{train}}(x_j)} \quad (24)$$

The decision rule for accepting or rejecting the null hypothesis is based on a threshold η for the test statistic.

$$\frac{\mathcal{H}_1}{\mathcal{H}_0} \geq \eta \quad (25)$$

The achieved false alarm rate of the detector, often called the False Positive Rate (FPR) in terms of classification, is defined as the fraction of normal points that are incorrectly labelled as anomalous. Results of the detector are interpreted in terms of the number of True Positives (TP), i.e. correct detections, False Positives (FP), i.e.

false alarms, True Negatives (TN), and False Negatives (FN). The FPR is given by the formula, $\text{FPR} = \text{FP}/(\text{FP} + \text{TN})$.

Interpreting this in the language of hypothesis testing, if we want a false alarm rate of α , we must ensure that the test statistic only exceeds the detection threshold $100 * \alpha\%$ of the time, under the null hypothesis. The test statistic CDF can be inverted at the $1 - \alpha$ quantile, to solve for the threshold ν that would give a FPR of α . The relationship between these is given by $\alpha = 1 - F(\eta)$, where the CDF of the test statistic is defined as F .

3.3.4. Bootstrap hypothesis testing

Our log-likelihood ratio hypothesis test does not produce a test statistic with an analytically tractable CDF, and therefore must be approximated using samples of observed data. Here, we use bootstrapping to approximate the CDF empirically with $\hat{F}$, Eq. (26), using bootstrap samples under the null hypothesis [19]:

$$\alpha = 1 - \hat{F}(\eta) = \frac{1}{B} \sum_{j=1}^B \mathbb{I}(\zeta^{(j)} > \eta) \quad (26)$$

where B is the number of bootstrap samples, $\zeta^{(j)}$ is the test statistic for the j^{th} bootstrap sample and $\mathbb{I}$ is the indicator function.

To generate the bootstrap samples, the training data $x_{0:K}$ is sampled with replacement B times, to give the collection of bootstrap test statistics $\{\zeta^{(j)}\}_{j=1}^B$. Since we are assuming the training data is free from anomalies, the assumption that the samples are taken under the null hypothesis is valid. Inverting the CDF in this case means solving Eq. (26) for η by ordering of the test statistics $\zeta^{(j)}$, and finding the $1 - \alpha$ quantile.

4. Experiments

The function approximation capabilities of the GPR for SDE coefficient estimation is demonstrated against polynomial regression. A known SDE is simulated from, and the diffusion coefficient estimated using both techniques, and the resulting approximating functions are compared against each other and the ground truth.

Our SDE-GPR anomaly detection algorithm (Algorithm 1) is evaluated with two examples. Firstly, testing is performed on a synthetic dataset generated to demonstrate the effectiveness of the SDE modelling approach for contextual anomalies and changes in underlying data generation dynamics. The second example is the NYC taxi dataset [20], which collects the total number of passengers within 30 minute intervals. This dataset has been used in a recent study of state-of-the-art univariate time-series anomaly

Algorithm 1: Anomaly Detection with SDE-GPR.

Input: Training data $x_{0:K}$, Test data $x_{K+1:M}$, Parameters: W, B, α, λ , Kernel functions: k_D, k ,

Output: Test data classifications

Training: Optimise marginal likelihood, Eq. (13) to set kernel hyperparameters (e.g. grid search)

Bootstrapping: Generate ζ_i for $x_{0:K}$.

Re-sample with replacement B times to find $\hat{F}$.

$\eta \leftarrow \hat{F}^{-1}(1 - \alpha)$

Testing:

for $i = K + 1 + W$ **to** M **do**

 Estimate $\mu_{\text{test}}(x)$ for $x_{i-W+1:i}$ using Eq. (16)

 Calculate ζ_i for $x_{i-W+1:i}$ using Eq. (21) **if** $\zeta_i > \eta$ **then**

x_i classified anomalous ;

else

x_i classified normal ;

end

Table 1
Parameter values for Synthetic and NYC experiments.

	SDE-GPR							ARIMA
Dataset	W	λ	B	training: k	training: k_D	test: k	test: k_D	(p,q,d)
Synthetic	40	0.01	10k	$k_{SE}, l = \frac{1}{4}, s = \frac{1}{5}$	$k_{SE}, l = 1, s = \frac{1}{5}$	$k_{poly}, p = 1, s = 1$	$k_{poly}, p = 0, s = \frac{1}{5}$	(1,0,0)
NYC	20	0.01	10k	$k_{SE}, l = \frac{1}{4}, s = \frac{1}{5}$	$k_{poly}, p = 0, s = \frac{1}{5}$	$k_{poly}, p = 1, s = 1$	$k_{poly}, p = 1, s = 1$	(1,1,2)

detection algorithms [3]. Present in this dataset are five labelled anomalies with known causes.

For comparison, a baseline method is tested alongside the SDE-GPR algorithm. This method assumes that the residual process is Gaussian distributed, and uses this to learn the mean and variance of the training dataset, i.e. $X_k \sim \mathcal{N}(\mu, \sigma^2)$. We can then set threshold analytically ($\eta = F^{-1}(\alpha/2) - \mu$) based on a desired false alarm rate using the Gaussian CDF $F(\cdot)$, declaring X_k an anomaly when $|X_k - \mu| > \eta$. Another more sophisticated time-series anomaly detection algorithm is used for comparison; the ARIMA(p, d, q) model [13] is fit to the data using maximum likelihood estimation, and optimal parameters are set by minimising the Bayesian information criteria (BIC) using a grid search over parameters. The squared one-step-ahead forecast error is used as the anomaly score, which is regarded as a classic method for anomaly detection [13]. The threshold is set empirically to the $1 - \alpha$ quantile of the ordered squared errors for the training dataset. See Table 1 for the parameters used for experimentation.

4.1. Synthetic bistable dataset

A bistable style SDE is used for analysis, in order to demonstrate the flexibility of GPR over more simplistic nonparametric regression methods. The form of the SDE is given below:

$$dX_t = -0.1X_t dt + \sqrt{2} \exp(-x^2/2) dW_t \quad (27)$$

See Section 5.1 for visual and numerical analysis of resulting function estimates given by GPR and polynomial regression.

4.2. Synthetic anomaly dataset

A synthetic dataset has been generated using a nonlinear AR model, with anomalous points ($t = [4800, 5600]$) in the test set generated by a linear AR(1) model. Total data points $N = 8000$, with first half used for training and the second for testing.

$$\text{Normal: } X_k \sim \mathcal{N}(0.1X_{k-1} - 0.3X_{k-1}^3, 0.15) \quad (28)$$

$$\text{Anomalous: } X_k \sim \mathcal{N}(0.8X_{k-1}, 0.1) \quad (29)$$

We use this to motivate the use of the SDE estimation for detecting changes to the underlying dynamics of data generation processes, that appear as contextual anomalies rather than the simpler, global case. Multiple realisations of the synthetic data are generated to assess the sensitivity of the performance metrics for each detector.

4.3. NYC Taxi dataset

For the NYC taxi data, a clear nonstationary seasonality can be observed, and so a moving average based seasonal, trend decomposition is used to extract the residuals of the process (`statsmodels.tsa.seasonal.seasonal_decompose`). Experiments used a train/test split of 30% / 70%. The detection thresholds have been set based on a desired false alarm rate of 1%. Weekly seasonality is assumed for the seasonal-trend decomposition.

5. Discussion & results

5.1. Function approximation

For the function approximation, an example realisation from the bistable SDE process is shown below, along with the polynomial, and Gaussian process regression results for the diffusion function. (See Fig. 1)

Fig. 2 shows the resulting mean functions learned using a GPR with SE kernel, $s = 1, l = 1$ and regularization $\lambda = 1$, and a polynomial regression order $n = 6$. The regularisation used for the Gaussian process regression provides a better behaved estimated function, with the function at the tails remaining close to zero, whereas the polynomial regression is not as well conditioned, and the function fluctuates significantly at the tails. To quantify the results, the mean squared error (MSE) between the ground truth function and the fitted functions are calculated for several polynomial orders and compared to the GPR fit; see Table 2 for results. Clearly at lower polynomial orders the estimated function is too simplistic to model the diffusion and therefore high MSE is observed, and likewise with very high polynomial orders the numerical instability and overfitting causes mismatch between the true diffusion and the estimated polynomial function. GPR outperformed the polynomial regression for all orders, which will be the case when the drift or diffusion functions are not easily captured by finite order polynomials.

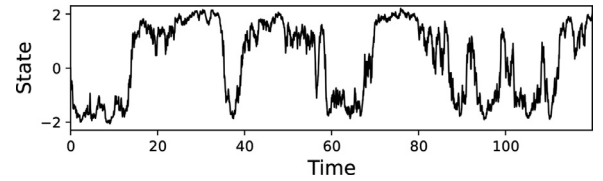


Fig. 1. Realisation of Bistable SDE.

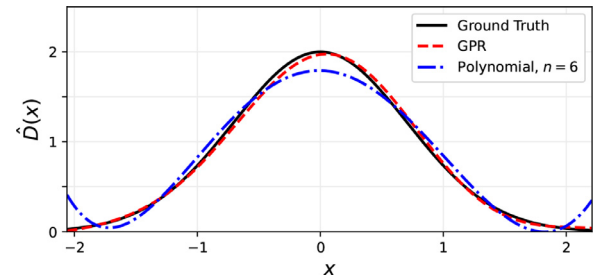


Fig. 2. Ground Truth diffusion function (black solid), GPR fit (red dashed) polynomial fit (blue dot-dashed).

Table 2
MSE of Approximation.

GPR	Polynomial				
	$n = 4$	$n = 6$	$n = 8$	$n = 15$	$n = 30$
0.0013	0.1176	0.0130	0.0027	0.0038	0.5260

Table 3
Synthetic results.

Detector	AUC	FPR	
		Desired	Realised
Baseline	0.595 ± 0.023	1.00%	$1.12\% \pm 0.39\%$
ARIMA	0.603 ± 0.026	1.00%	$1.12\% \pm 0.37\%$
SDE-GPR	0.976 ± 0.086	1.00%	$1.70\% \pm 0.86\%$

Table 4
NYC taxi results.

Detector	AUC	FPR		
		Desired	Realised	Corrected
Baseline	0.887	1.00%	5.63%	1.42%
ARIMA	0.697	1.00%	1.44%	0.46%
SDE-GPR	0.963	1.00%	5.80%	1.85%

5.2. Anomaly detection

In order to evaluate the performance of the detectors, the area under the receiver operator characteristic curve (AUC) will be calculated, by varying the threshold and monitoring the TPR and FPR. An AUC score of 1 is a perfect classifier, and a score of 0.5 is comparable to random classification. Results for the AUC scores along with the comparison between desired and realised FPR are given in Table 3 for the synthetic, and Table 4 for the NYC taxi data for each detector, which allows us to compare the threshold setting methods used (analytic for the baseline, empirical for ARIMA, and bootstrapping for SDE-GPR).

5.2.1. Synthetic anomaly results

The SDE-GPR detector significantly outperformed the baseline and ARIMA approach on the synthetic dataset with the contextual anomalies (see Fig. 3), and provided a near perfect AUC. The baseline detector was unable to distinguish between the anomalies and the normal model because the anomalous change was in the dynamics of the data generation, and was not apparent when purely considering the size of deviations from the mean, shown by the AUC being close to random classification. For ARIMA, the learned model was AR(1), which is a linear regression on the previous data point, however the true model is a nonlinear regression and so the ARIMA model is unable to capture the dynamics and performed poorly in terms of AUC compared to SDE-GPR.

The realised FPR for the baseline detector closely matched the desired FPR for the synthetic dataset, which is due to the normal model producing samples that are similar in distribution to a Gaussian PDF, and so the analytical approach to setting the threshold worked well. This closely matched FPR is also observed for

ARIMA, where residuals are assumed to be Gaussian when fitting the model. SDE-GPR resulted in a well matched, but slightly inflated FPR compared to desired, which is due to the test sliding window effect causing a delay in initial anomaly detection, and previous anomalous samples appearing in the sliding window after the anomaly period ends.

5.2.2. NYC Taxi results

For the NYC taxi data, SDE-GPR again outperformed the baseline and ARIMA detectors, with a near perfect AUC. This performance is a result of SDE-GPR's ability to better suppress the anomaly score when observing test data that is similar to the training model, and providing better discrimination of anomalous behaviour compared to the baseline that relies on the size of residuals alone. It is worth noting that both the baseline and ARIMA models produced a high AUC despite their simplicity, which is a result of the detrending and deseasonalizing preprocessing steps that extracted the residuals; all anomalies were clearly present in the residual process, which highlights the importance of detrending and deseasonalizing periodic data when attempting to detect anomalies. Note the ARIMA model (5,1,2) produces a much better AUC = 0.881, but the BIC optimisation favours a model (1,1,2) with AUC = 0.697 due to its simplicity, and in this case where training data is abundant and overfitting less likely, a prediction MSE may be a better metric for model selection. The NYC dataset includes only five individual time points labelled as anomalies, but the data displays anomalous behaviour at many more time points surrounding these labels, which results in artificially inflated FPRs compared to desired (see Table 4). In order to correct for this and record a more realistic FPR, we include a 'corrected' FPR, which only counts the false positives outside the regions where these anomalies have impacted the data (see Fig. 4 for correctly labelled false alarms). After this correction, we observe similar results to the synthetic realised FPR, which shows SDE-GPR and the baseline controlling the FPR to closely match the desired, with SDE-GPR fractionally over-detecting compared to the baseline due to the sliding window effect. The ARIMA model produced realised and corrected FPR either side of the desired value. For practical applications, these effects will have to be considered; the sliding window size must be set based on expected anomaly duration, and how tightly controlled the false alarm rate should be, and would vary with application. In practice, false alarm rates will be set to low values, particularly if data is received with high frequency, therefore, the main focus of estimating the test statistic PDF is in the tail of the distribution. Extensions to the simple bootstrapping method could improve these tail estimates, for example, the smoothed bootstrap may provide more robust estimates of $1 - \alpha$ quantiles.

6. Conclusion

Our proposed method SDE-GPR for anomaly detection is shown to distinguish between normal data generation and anomalous behaviour through automatic comparison of underlying dynamics. SDE-GPR outperformed a baseline method of detection for both synthetic and NYC taxi datasets in terms discriminative power, whilst still providing control over false alarm rates. SDE-GPR also discriminates contextual anomalies better compared to the ARIMA model based on AUC. Possible future work could include adaptation to nonstationary conditions, or extension of bootstrapping methods to further improve the setting of detection thresholds. In this work, the nonstationary, periodic data was detrended and deseasonalized, with the aim of providing a stationary residual process to model. However, nonstationarity may well be present in the residuals, which can, for example, be observed in the increased variation of taxi rides in January compared to training data.

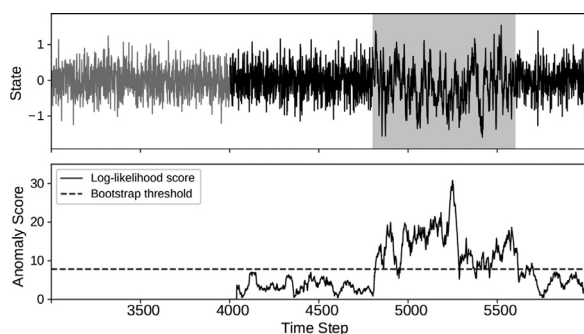


Fig. 3. Synthetic dataset results. Above: Synthetic training data (grey) and test data (black) with anomalous data (shaded). Below: Log-likelihood score from SDE-GPR (solid black) with bootstrap threshold (dashed black).

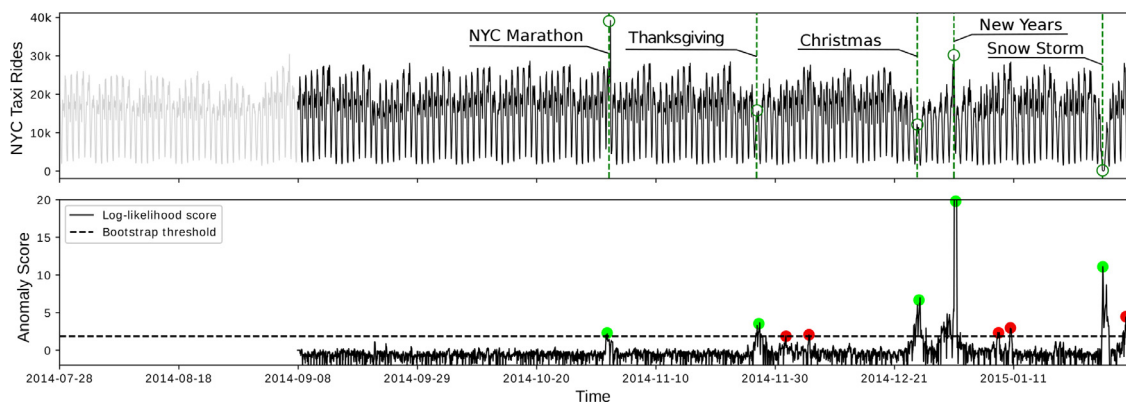


Fig. 4. NYC taxi dataset results. Above: NYC Taxi training data (grey) and Test data (black) with known anomalies and causes. Below: Log-likelihood score from SDE-GPR (solid black) with bootstrap threshold (dashed black). Detections highlighted with dots, known anomalies detected (green) and false alarms (red). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Further work could look at adapting the normal model to non-stationarity in the data, either passively using a sliding window approach, or actively reconstructing a normal model when significant change to standard data generation is observed. Additional to this, work could focus on improving the algorithmic efficiency of the SDE estimation, as the naïve GPR implementation has $\mathcal{O}(N^3)$ time complexity, which would become infeasible for high-frequency data sources. One example of this would be the Nyström method for reducing this to $\mathcal{O}(Nk^2)$, $k \ll N$ using subsampling techniques. Function approximation results demonstrate the flexibility of the GPR, and its robustness to overfitting through regularization when compared to polynomial regression.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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